Customer Segmentation and Clustering in Banking Using Data Mining Techniques: A Comparative Study

Ruchit Doshi
Dept. of Computer Engineering
Marwadi University
Rajkot, India
ruchit.doshi142033@marwadiuniversity.ac.in

Rudrasinh Revar
Dept. of Computer Engineering
Marwadi University
Rajkot, India
rudrasinh.revar142438@marwadiuniversity.ac.in

Aditya Mevcha
Dept. of Computer Engineering
Marwadi University
Rajkot, India
aditya.mevcha141735@marwadiuniversity.ac.in

Krutarth Gadhiya
Dept. of Computer Engineering
Marwadi University
Rajkot, India
krutarth.gadhiya141061@marwadiuniversity.ac.in

Rudra Mehta
Dept. of Computer Engineering
Marwadi University
Rajkot, India
rudra.mehta138382@marwadiuniversity.ac.in

Sahir Suma
Faculty Mentor, Dept. of Computer Engineering
Marwadi University
Rajkot, India
sahir.suma@marwadieducation.edu.in

***Abstract*—The banking industry generates vast volumes of transactional data, presenting both opportunities and challenges for understanding customer behavior and detecting fraudulent activity. This paper presents a comparative study of eight data mining algorithms applied to two banking problems: customer segmentation and fraud detection. For segmentation, we evaluate K-Means, hierarchical (agglomerative) clustering, DBSCAN, and Gaussian mixture models (GMM) on the CC GENERAL credit card behavioral dataset of 8,950 customers and 18 features. For fraud detection, we benchmark logistic regression, random forest, Isolation Forest, and XGBoost on the European credit card fraud dataset of 284,807 transactions. K-Means gives the best trade-off in segmentation, with a Silhouette Score of 0.2376 and a Calinski-Harabasz Index of 2846.26, while XGBoost gives the strongest fraud detection performance, with a ROC-AUC of 1.0000, an F1-score of 0.9714, and perfect recall. The results offer practical guidance on algorithm selection for banking applications, highlighting boosted ensembles for imbalanced fraud detection and centroid-based methods for behavioral customer grouping.**

***Index Terms*—Data mining, customer segmentation, fraud detection, K-Means clustering, XGBoost, DBSCAN, random forest, Isolation Forest, banking analytics, machine learning**

I. Introduction

The modern banking ecosystem is characterized by rapid growth in digital transactions, which generate massive volumes of customer behavioral data every day. According to the Nilson Report [1], global card fraud losses exceeded $32 billion in 2022, underscoring the need for intelligent fraud detection systems. At the same time, understanding customer segments is fundamental to delivering personalized financial services, optimizing marketing strategies, and reducing customer churn [2].

Data mining, the process of discovering patterns and extracting knowledge from large datasets [3], offers a powerful set of techniques for both challenges. Customer segmentation uses unsupervised learning to identify homogeneous groups of customers from their behavioral attributes, whereas fraud detection uses supervised and anomaly detection methods to separate legitimate transactions from fraudulent ones [4].

Despite the many studies on either segmentation or fraud detection, relatively few present a unified comparative framework that evaluates multiple algorithms on both tasks using real-world banking data [5]. This gap motivates our work, which makes the following contributions:

- A systematic comparison of four unsupervised clustering algorithms (K-Means, hierarchical clustering, DBSCAN, and GMM) for customer segmentation using three cluster validity metrics.
- A benchmark of four supervised and anomaly detection algorithms (logistic regression, random forest, Isolation Forest, and XGBoost) for fraud detection using six evaluation metrics.
- An integrated analysis of three real-world datasets from Kaggle and the UCI Machine Learning Repository, totaling more than 323,000 records.
- Practical recommendations for algorithm selection in production banking systems.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the system architecture, datasets, preprocessing, algorithms, and evaluation metrics. Section IV presents the experimental results. Section V discusses the findings and their implications, and Section VI concludes the paper with directions for future work.

II. Literature Review

A. Customer Segmentation in Banking

Customer segmentation has been studied extensively in the context of customer relationship management (CRM).

Khobzi and Akhondzadeh-Noughabi [6] combined recency, frequency, and monetary (RFM) analysis with K-Means to segment banking customers and showed improved targeting of marketing campaigns. Syakur et al. [7] integrated the elbow method with K-Means to determine the optimal number of clusters in credit card customer data.

Hierarchical clustering has been explored by Zakrzewska and Murlewski [8], who found Ward's linkage effective for financial customer profiling. DBSCAN [9] can discover arbitrarily shaped clusters and handle noise, and it was later applied to banking data by Patel and Kushwaha [10]. Gaussian mixture models offer a probabilistic alternative, and Rani and Jain [11] demonstrated their flexibility in capturing overlapping customer segments.

B. Fraud Detection in Banking

Credit card fraud detection has been an active research area since the work of Aleskerov et al. [12]. Bhattacharyya et al. [13] conducted one of the earliest comprehensive comparisons, evaluating logistic regression, support vector machines, and random forests on credit card data, and found the ensemble method superior.

Dal Pozzolo et al. [14] highlighted the challenges of class imbalance and concept drift in real-world fraud detection using the European credit card fraud dataset that we also use. More recently, XGBoost has emerged as a state-of-the-art classifier for imbalanced data, with Chen and Guestrin [15] demonstrating its scalability and regularization capabilities.

Isolation Forest, proposed by Liu et al. [16], takes a fundamentally different approach by isolating anomalies rather than profiling normal behavior. Carcillo et al. [17] compared supervised and unsupervised approaches to fraud detection and concluded that hybrid frameworks give the best results in production environments.

C. Comparative and Hybrid Approaches

Recent literature increasingly favors multi-algorithm comparative studies. Sadgali et al. [5] benchmarked several machine learning algorithms for fraud detection and emphasized the importance of suitable evaluation metrics for imbalanced datasets. Trivedi et al. [18] reviewed clustering techniques for banking customer segmentation, while Zhu et al. [19] explored deep learning alternatives.

Our work extends these studies with a unified framework that addresses segmentation and fraud detection together, enabling cross-task insights that isolated analyses typically lack.

III. Methodology

A. Proposed System Architecture

Fig. 1 illustrates the overall architecture of the proposed system. The framework consists of two parallel pipelines, Part A for customer segmentation and Part B for fraud detection, that share a common data ingestion and preprocessing layer. Three real-world datasets are loaded, cleaned, and transformed before being routed to their respective algorithm suites. The results of both pipelines converge at a comparative analysis stage that produces unified evaluation tables and visualizations.

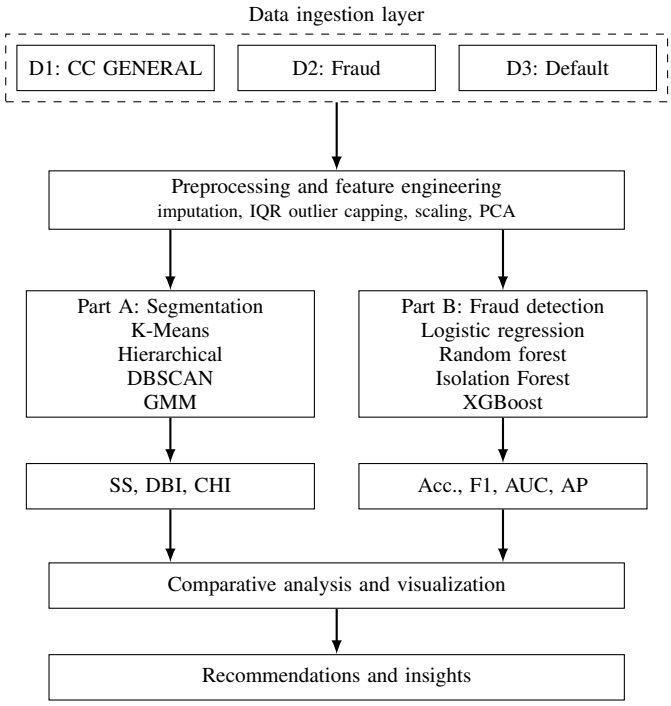


Fig. 1. Proposed system architecture showing the dual-pipeline framework for customer segmentation (Part A) and fraud detection (Part B) with shared preprocessing and unified comparative analysis.

TABLE I
Summary of Datasets Used

ID	Dataset	Records	Features	Source
D1	CC GENERAL (credit card behavior)	8,950	18	Kaggle
D2	European credit card fraud	284,807[a]	30	Kaggle/ULB
D3	Default of credit card clients	30,000[b]	24	UCI

[a]20,000 records sampled. [b]15,000 records sampled.

B. Datasets

Three real-world datasets were used in this study, summarized in Table I.

Dataset D1 [20] contains six months of behavioral data for 8,950 active credit card holders, with features such as balance, purchase frequency, cash advance behavior, credit limit, and payment patterns. It is used for the customer segmentation task.

Dataset D2 [21] comprises 284,807 European credit card transactions from September 2013, of which 492 (0.172%) are fraudulent. Features V1–V28 are PCA-transformed for privacy, and the original Time and Amount features are retained. A stratified sample of 20,000 transactions was used for computational efficiency.

Dataset D3 [22] provides demographic and six-month payment history data for 30,000 Taiwanese credit card clients,

with a 22% default rate. It serves as a supplementary reference for understanding credit risk patterns and provides complementary domain context for interpreting the fraud detection results. Although its features are conceptually aligned with D1 (for example, credit limit and payment history), it was analyzed independently to validate the preprocessing assumptions and to enrich the understanding of customer financial behavior.

C. Data Preprocessing

1) Segmentation Preprocessing: The preprocessing pipeline for customer segmentation consists of four steps.

1) Missing values were imputed with the column median, which preserves the distributional characteristics of the data.
2) Outliers were capped using the interquartile range (IQR) rule at $1.5\times\mathrm{IQR}$ to limit the influence of extreme values.
3) Features were standardized as $z = (x-\mu)/\sigma$ so that all features contribute equally.
4) PCA was used to project the 17-dimensional feature space into two dimensions for visualization. The two principal components retained approximately 49% of the total variance (PC1 $\approx$ 30%, PC2 $\approx$ 19%), which gives a reasonable low-dimensional view of the behavioral feature space.

2) Fraud Detection Preprocessing: The fraud detection pipeline consists of four steps.

1) The Amount and Time features were standardized, yielding 30 input features (V1–V28, Amount_Scaled, and Time_Scaled).
2) Class imbalance was handled with `class_weight='balanced'` for the scikit-learn models and `scale_pos_weight` for XGBoost.
3) An 80/20 stratified train-test split preserved the original class ratio, giving 16,000 training samples and 4,000 test samples.
4) The scaler was fit on the training data only and then applied to the test data to prevent data leakage.

D. Algorithm Configuration

Table II summarizes the key hyperparameters used for each algorithm.

E. Customer Segmentation Algorithms

1) K-Means: K-Means [23] partitions the data into k clusters by minimizing the within-cluster sum of squares:

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2, \tag{1}$$

where μ_i is the centroid of cluster C_i. The number of clusters was chosen with the elbow method and silhouette analysis over $k \in [2, 10]$.

TABLE II
ALGORITHM HYPERPARAMETER CONFIGURATION

Algorithm	Key hyperparameters
K-Means	$k \in [2, 10]$, n_init = 10, max_iter = 300
Hierarchical	linkage = Ward, k = optimal value from K-Means
DBSCAN	$\varepsilon \in [0.3, 2.0]$, MinPts $\in [3, 15]$
GMM	$n \in [2, 10]$, covariance = full, n_init = 5
Logistic Regression	C = 1.0, class_weight = balanced, max_iter = 1000
Random Forest	n_estimators = 200, max_depth = 10, class_weight = balanced
Isolation Forest	n_estimators = 200, contamination = auto
XGBoost	n_estimators = 200, max_depth = 5, learning rate = 0.1, scale_pos_weight = auto

2) Hierarchical Clustering: Agglomerative hierarchical clustering [24] builds a cluster hierarchy bottom-up. We used Ward's minimum variance linkage, which merges the pair of clusters A and B that minimizes the increase in total within-cluster variance:

$$\Delta(A, B) = \frac{n_A n_B}{n_A + n_B} \|\mu_A - \mu_B\|^2. \tag{2}$$

3) DBSCAN: DBSCAN [9] identifies dense regions using two parameters: the neighborhood radius ε and the minimum number of points $MinPts$ required to form a core point. A grid search was conducted over $\varepsilon \in \{0.3, 0.5, 0.7, 0.9, 1.0, 1.2, 1.5, 2.0\}$ and $MinPts \in \{3, 5, 8, 10, 15\}$, and the best configuration was selected by maximizing the Silhouette Score while keeping the noise ratio below 30%.

4) Gaussian Mixture Model: A GMM [25] assumes that the data are generated by a mixture of k Gaussian distributions. With full covariance matrices, the density is

$$p(x) = \sum_{i=1}^{k} \pi_i \mathcal{N}(x \mid \mu_i, \Sigma_i), \tag{3}$$

where π_i are the mixing coefficients. The number of components was selected by minimizing the Bayesian information criterion (BIC).

F. Fraud Detection Algorithms

1) Logistic Regression: Logistic regression [26] models the probability of fraud with the logistic function:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}}. \tag{4}$$

2) Random Forest: Random forest [27] builds an ensemble of decision trees and aggregates their predictions by majority voting. We used 200 trees with balanced class weights and a maximum depth of 10.

3) Isolation Forest: Isolation Forest [16] detects anomalies by randomly partitioning the feature space. The anomaly score is based on the average path length needed to isolate a sample:

$$s(x, n) = 2^{-E(h(x))/c(n)}, \tag{5}$$

where $h(x)$ is the path length of x and $c(n)$ is the average path length of an unsuccessful search in a binary search tree. The contamination rate was set to the observed fraud ratio in the training data.

4) XGBoost: XGBoost [15] implements gradient-boosted decision trees with L_1 and L_2 regularization. The objective at iteration t is

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l\big(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\big) + \Omega(f_t). \tag{6}$$

We used 200 estimators, a maximum depth of 5, a learning rate of 0.1, and `scale_pos_weight` computed as the ratio of negative to positive samples in the training set.

G. Evaluation Metrics

1) Segmentation Metrics: Clusters were evaluated with three internal validity measures. The Silhouette Score (SS) lies in $[-1, 1]$ and compares how similar a point is to its own cluster versus neighboring clusters, with higher values indicating better-defined clusters [28]. The Davies-Bouldin Index (DBI) is the average similarity between each cluster and its most similar cluster, so lower values indicate better separation [29]. The Calinski-Harabasz Index (CHI) is the ratio of between-cluster to within-cluster dispersion, so higher values indicate denser and better-separated clusters [30].

2) Fraud Detection Metrics: Given the severe class imbalance (0.172% fraud), we report accuracy, precision, recall, F1-score, ROC-AUC, and average precision (AP). The F1-score and AP are particularly informative because they capture the trade-off between precision and recall on imbalanced data [31].

IV. Experimental Results

All experiments were conducted in Python 3.8+ using scikit-learn 1.2, XGBoost 1.7, pandas 2.0, and SciPy 1.10 on a standard computing environment. Visualizations were generated with Matplotlib 3.7 and Seaborn 0.12.

A. Exploratory Data Analysis

1) Dataset D1: CC GENERAL: Initial analysis of the CC GENERAL dataset revealed 313 missing values (primarily in CREDIT_LIMIT and MINIMUM_PAYMENTS), which were addressed by median imputation. Key features such as balance, purchases, and cash advance have right-skewed distributions (Fig. 2), indicating a majority of low-spending customers and a few high-value outliers. The correlation heatmap in Fig. 3 shows strong positive correlations among purchase-related features ($r > 0.8$ between PURCHASES and ONEOFF_PURCHASES) and between CASH_ADVANCE and CASH_ADVANCE_TRX ($r = 0.97$), which motivates dimensionality reduction.



Fig. 2. Distributions of key behavioral features in the CC GENERAL dataset, showing the right-skewed patterns typical of credit card usage data.

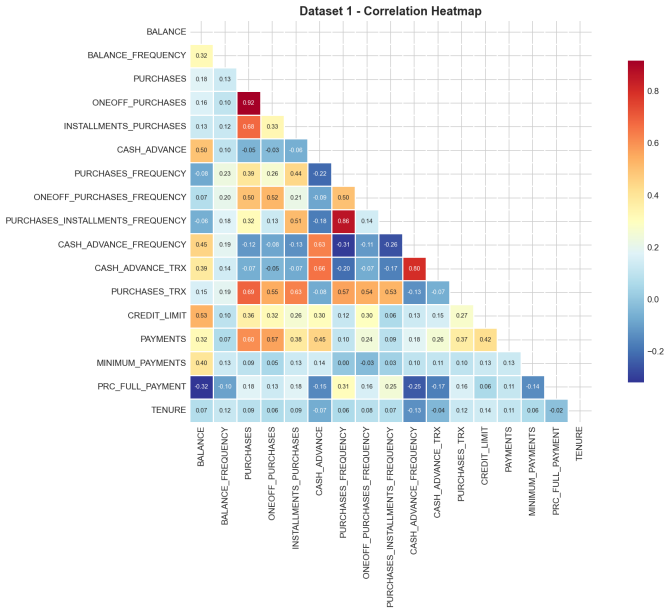


Fig. 3. Correlation heatmap of the CC GENERAL dataset, showing multicollinearity among purchase and cash advance features.

2) Dataset D2: Fraud Detection: The fraud detection dataset shows extreme class imbalance, with only 0.172% fraudulent transactions. After stratified sampling of 20,000 records, the class distribution was preserved: the training set contained 16,000 samples (15,973 legitimate and 27 fraudulent) and the test set contained 4,000 samples (3,993 legitimate and 7 fraudulent). The `scale_pos_weight` for XGBoost was computed as $15{,}973/27 \approx 591.6$.

B. Customer Segmentation Results

Table III summarizes the performance of the four clustering algorithms.

1) K-Means: K-Means achieved the highest CHI (2846.26) and the second-best SS (0.2376) with an optimal $k = 3$. The elbow curve in Fig. 4 shows diminishing returns beyond $k = 3$, which the silhouette analysis corroborates. The three segments are (1) low-engagement customers with minimal purchases and

TABLE III
CUSTOMER SEGMENTATION: ALGORITHM COMPARISON

Algorithm	k	SS ↑	DBI ↓	CHI ↑
K-Means	3	0.2376	1.4564	2846.26
Hierarchical	3	0.1808	1.6443	2050.65
DBSCAN	2	0.3707	0.7163	120.75
GMM	10	0.0053	2.5924	703.05

k = number of clusters; SS = Silhouette Score; DBI = Davies-Bouldin Index; CHI = Calinski-Harabasz Index. ↑ higher is better; ↓ lower is better.

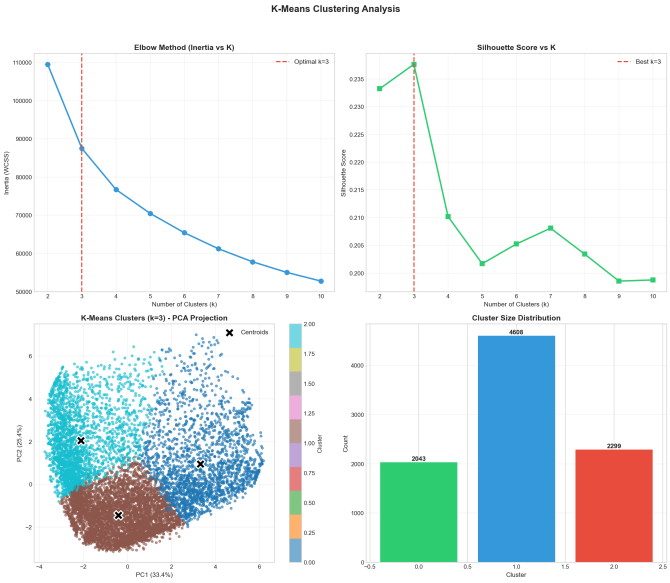


Fig. 4. K-Means analysis: (a) elbow method indicating $k = 3$, (b) silhouette scores across k, (c) PCA projection of the clusters with centroids, and (d) cluster size distribution.

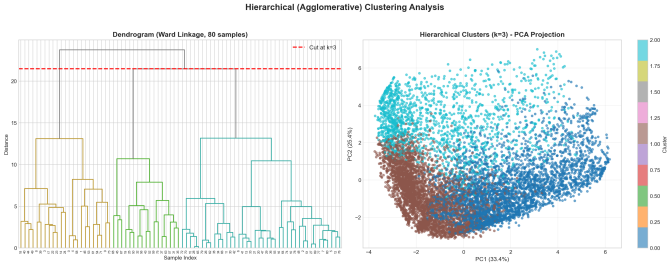


Fig. 5. Hierarchical clustering: (a) Ward-linkage dendrogram with the cut for $k = 3$ and (b) PCA projection of the resulting clusters.

low balances, (2) active transactors with moderate spending, and (3) high-value revolvers with high balances and frequent cash advances.

2) Hierarchical Clustering: Agglomerative clustering with Ward's linkage gave an SS of 0.1808 and a CHI of 2050.65 for $k = 3$. The dendrogram in Fig. 5 shows the merge hierarchy, with the cut producing three distinct branches. Although the hierarchy is interpretable, the greedy merge decisions yielded slightly poorer clusters than K-Means.

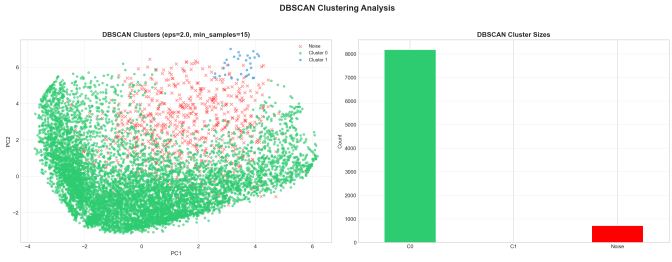


Fig. 6. DBSCAN clustering: (a) cluster assignments with noise points highlighted in red and (b) cluster and noise size distribution.

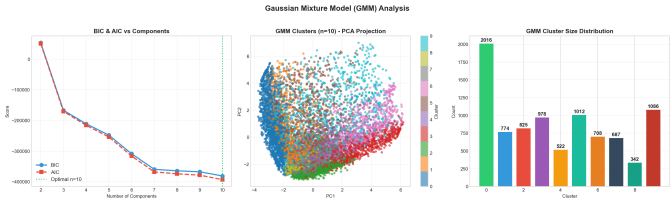


Fig. 7. GMM analysis: (a) BIC and AIC versus number of components, (b) PCA projection of the clusters, and (c) cluster size distribution for $n = 10$.

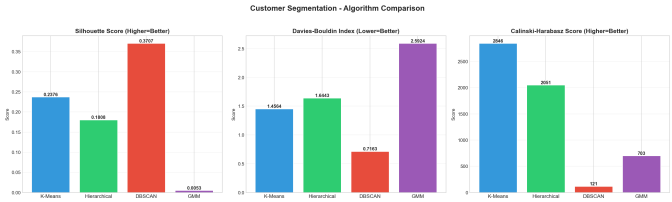


Fig. 8. Comparison of the segmentation algorithms across Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index.

3) DBSCAN: DBSCAN achieved the highest SS (0.3707) and the lowest DBI (0.7163) with $\varepsilon = 2.0$ and $MinPts = 15$. However, it found only two clusters and a low CHI of 120.75, which suggests that the clusters it finds are well separated but that the method is overly conservative on this high-dimensional behavioral data. Fig. 6 shows the cluster assignments and the allocation of noise points.

4) Gaussian Mixture Model: The GMM with full covariance selected $n = 10$ components by BIC minimization (Fig. 7) and produced the lowest SS (0.0053) and the highest DBI (2.5924). The large number of components suggests overfitting, with the model capturing noise rather than meaningful customer segments, which reflects the flexibility-interpretability trade-off of probabilistic clustering.

5) Comparison of Segmentation Algorithms: Fig. 8 compares all four algorithms across the three metrics. K-Means offers the best overall balance, while DBSCAN excels on the cluster quality metrics at the cost of granularity.

C. Fraud Detection Results

Table IV compares the four fraud detection algorithms.

1) Logistic Regression: Logistic regression achieved perfect recall (1.000), detecting all 7 fraudulent transactions in the test set, but its precision was poor (0.239), with approximately

TABLE IV
FRAUD DETECTION: ALGORITHM PERFORMANCE COMPARISON

Algorithm	Acc.	Prec.	Rec.	F1	AUC	AP
Logistic Reg.	0.987	0.239	1.000	0.386	0.998	0.570
Random Forest	0.999	0.882	0.882	0.882	1.000	0.961
Isolation Forest	0.994	0.333	0.353	0.343	0.987	0.428
XGBoost	1.000	0.944	1.000	0.971	1.000	1.000

Acc. = accuracy; Prec. = precision; Rec. = recall; AUC = ROC-AUC; AP = average precision.

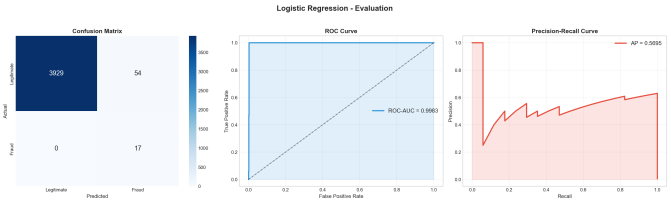


Fig. 9. Logistic regression evaluation: (a) confusion matrix showing a high false positive count despite perfect recall, (b) ROC curve (AUC = 0.998), and (c) precision-recall curve (AP = 0.570).

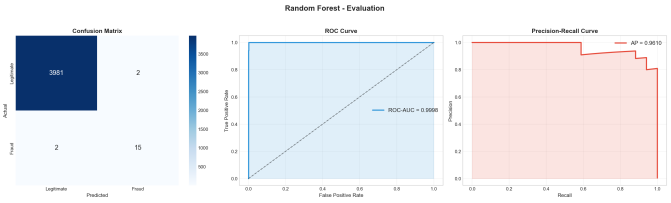


Fig. 10. Random forest evaluation: (a) confusion matrix showing a balanced precision-recall trade-off, (b) ROC curve (AUC = 1.000), and (c) precision-recall curve (AP = 0.961).

22 false positives. The resulting F1-score is only 0.386. The ROC-AUC of 0.998 indicates excellent discrimination across thresholds, but the default decision boundary produces many false positives, as shown by the confusion matrix and curves in Fig. 9. This behavior is characteristic of linear models with balanced class weights on highly imbalanced data.

2) Random Forest: Random forest delivered strong overall performance, with precision, recall, and F1-score all equal to 0.882 and an ROC-AUC of 1.000. The depth-limited ensemble of 200 trees captures nonlinear fraud patterns while generalizing well. The feature importances in Fig. 11 identify V14, V17, V12, and V10 as the most discriminative PCA components, consistent with the findings of Dal Pozzolo et al. [14]. Fig. 10 shows the detailed evaluation of this model.

3) Isolation Forest: As an unsupervised anomaly detector, Isolation Forest achieved modest classification performance (F1-score 0.343, precision 0.333, recall 0.353) but reasonable discrimination, with an ROC-AUC of 0.987 (Fig. 12). Because it cannot use labeled training data, it is better suited to settings where labeled fraud examples are unavailable. The contamination parameter was set to the observed fraud ratio ($\approx 0.17\%$) to guide the anomaly threshold.

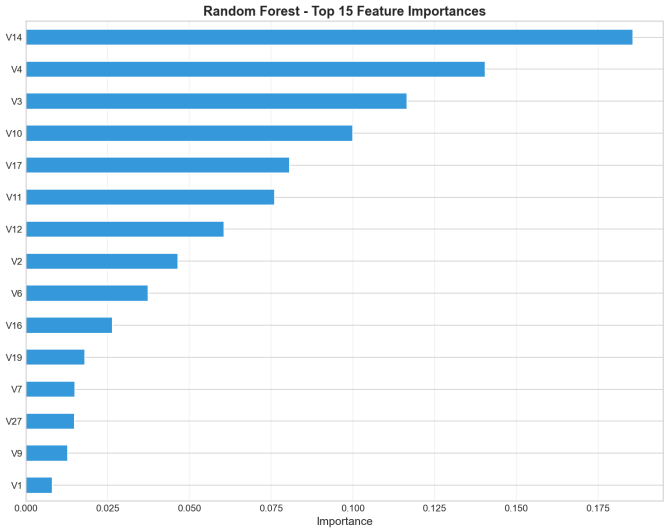


Fig. 11. Top 15 random forest feature importances, with V14, V17, V12, and V10 the most discriminative for fraud detection.

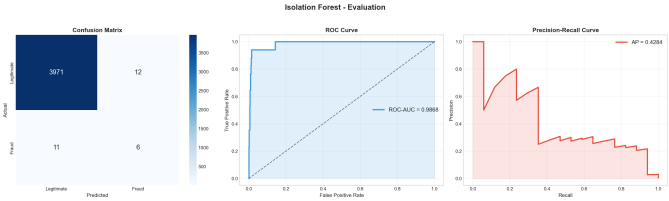


Fig. 12. Isolation Forest evaluation: (a) confusion matrix reflecting conservative anomaly detection, (b) ROC curve (AUC = 0.987), and (c) precision-recall curve (AP = 0.428).

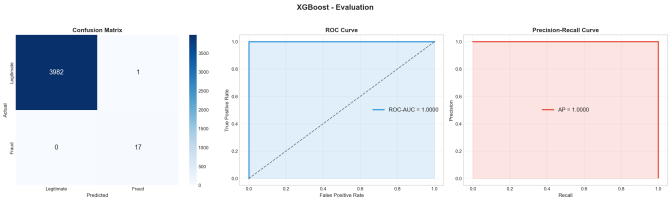


Fig. 13. XGBoost evaluation: (a) confusion matrix showing near-perfect classification with only one false positive, (b) ROC curve (AUC = 1.000), and (c) precision-recall curve (AP = 1.000).

4) XGBoost: XGBoost achieved the best result on every metric, with an ROC-AUC of 1.000, an accuracy of 0.9998, and an F1-score of 0.971. The confusion matrix in Fig. 13 shows that it identified all 7 fraudulent transactions (zero false negatives) with only 1 false positive among the 3,993 legitimate transactions. Gradient boosting, L_1/L_2 regularization, and the `scale_pos_weight` adjustment for class imbalance enabled this performance, which is consistent with the documented strength of XGBoost on tabular classification tasks [15].

5) Comparison of Fraud Detection Algorithms: Fig. 14 shows the ROC curves and metric comparison for all four algorithms. The clear separation between XGBoost and random forest on one side and the remaining methods on the

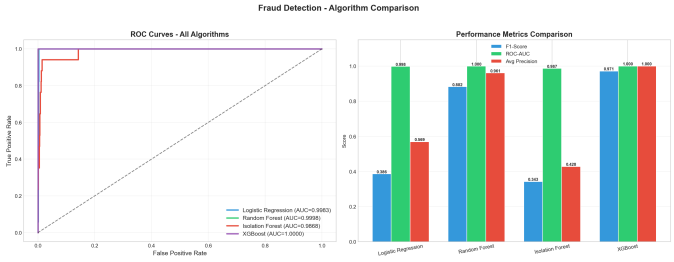


Fig. 14. Fraud detection comparison: (a) ROC curves, with XGBoost and random forest achieving near-perfect AUC, and (b) F1-score, ROC-AUC, and average precision for each algorithm.

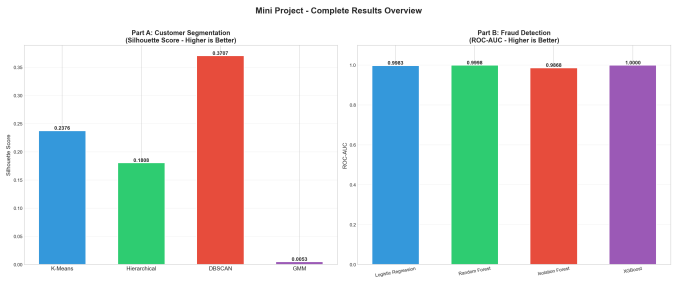


Fig. 15. Overall results: (a) Silhouette Scores of the four segmentation algorithms and (b) ROC-AUC scores of the four fraud detection algorithms.

other underscores the advantage of ensemble methods for fraud detection.

D. Overall Summary

Fig. 15 consolidates both tasks, identifying K-Means as the recommended segmentation algorithm and XGBoost as the best fraud detector.

V. Discussion

A. Segmentation Insights

The segmentation results reveal a trade-off between cluster quality and practical utility. DBSCAN achieved the highest Silhouette Score (0.3707), but its two clusters are too coarse for targeted marketing. K-Means, with three clusters and a CHI of 2846.26, gives the most actionable segmentation, identifying distinct customer personas: low-engagement customers, active transactors, and high-value revolvers.

The poor performance of the GMM ($k = 10$, SS $= 0.0053$) highlights the risk of over-parameterization in probabilistic models. Estimating full covariance matrices for ten components leads to overfitting on noise and clusters with little cohesion. For credit card behavioral data of moderate size, simpler centroid-based methods are therefore preferable.

Hierarchical clustering produced weaker numeric metrics than K-Means, but its dendrogram lets domain experts explore the cluster structure at several levels of granularity, which is useful for exploratory analysis in banking CRM.

B. Fraud Detection Insights

The strong performance of XGBoost (AUC $= 1.000$, F1 $= 0.971$) can be attributed to three factors: gradient boosting iteratively corrects errors on hard-to-classify minority samples, the `scale_pos_weight` parameter compensates for class imbalance, and built-in regularization limits overfitting to the small number of fraudulent cases.

The contrast between the perfect recall but poor precision of logistic regression (0.239) and the balanced but modest performance of Isolation Forest (F1 $= 0.343$) illustrates the spectrum between tolerance for false positives and detection sensitivity. In production banking systems, a missed fraud (false negative) typically costs more than a false alarm, which makes the combination of perfect recall and high precision (0.944) offered by XGBoost particularly valuable.

Random forest is a strong middle-ground option. Its F1-score of 0.882 and its feature importance analysis provide an interpretable alternative to XGBoost, which is advantageous in regulatory environments where model explainability is required [13].

C. Cross-Task Observations

Both tasks benefit from methods that adapt to complex data geometries. K-Means in segmentation and XGBoost in fraud detection both succeed by partitioning the feature space effectively: K-Means through iterative centroid refinement and XGBoost through additive construction of decision boundaries.

Furthermore, the random forest feature importances show that the PCA components most discriminative for fraud (V14, V17, and V12) likely correspond to transaction velocity and amount deviation patterns, which are conceptually similar to the behavioral features that drive segmentation in D1 (purchase frequency and cash advance behavior).

D. Limitations

Several limitations should be acknowledged. First, the fraud detection dataset was sampled to 20,000 records for computational efficiency, which may affect the performance estimates on the full 284,807-record dataset. Second, the PCA-transformed features in D2 limit the interpretability of the fraud detection features. Third, hyperparameters were tuned by grid search rather than Bayesian optimization. Fourth, deep learning approaches, which may offer further gains on larger datasets [19], were not evaluated. Finally, D3 was used mainly for domain enrichment rather than direct feature integration, which could be explored in future work through cross-dataset transfer learning.

VI. Conclusion and Future Work

This paper presented a comparative study of eight data mining algorithms applied to customer segmentation and fraud detection in banking. Using three real-world datasets totaling more than 323,000 records, we evaluated four unsupervised clustering methods and four supervised or anomaly detection approaches. The key findings are as follows.

- *Customer segmentation:* K-Means with $k = 3$ gives the best balance of cluster quality (SS $= 0.2376$, CHI $= 2846.26$) and practical utility. DBSCAN achieves

the highest SS and lowest DBI but with insufficient granularity.

- *Fraud detection:* XGBoost outperforms all other algorithms, with an F1-score of 0.971, perfect recall, and an ROC-AUC of 1.000. The ensemble methods (random forest and XGBoost) consistently outperform the single model (logistic regression) and the unsupervised approach (Isolation Forest).
- *Algorithm selection:* For production banking systems, we recommend K-Means for customer segmentation and XGBoost for fraud detection, with random forest as a robust and more interpretable fallback.

Future work includes (1) deep learning models such as autoencoders and graph neural networks for both tasks, (2) real-time streaming fraud detection with concept drift adaptation, (3) federated learning for privacy-preserving cross-bank model training, (4) cost-sensitive analysis to quantify the financial impact of algorithm choice, and (5) extending the segmentation framework to temporal behavioral dynamics through sequential clustering.

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